Subscription Churn & Revenue Analysis

Case study inspired by Netflix's 2022 subscriber-loss event | SQL · Excel · Power BI

Objective

In Q1 2022, Netflix reported its first subscriber loss in a decade, wiping 35% off its stock price in a single day and putting subscription-business churn under global scrutiny. This project analyzes a subscription customer base (public proxy dataset, since real customer-level data is not publicly available) to identify which customer segments carry the highest churn risk, quantify the revenue exposure, and recommend retention actions.

Methodology

Churn was defined in SQL as no payment recorded within 30 days of the dataset's snapshot date. Six candidate drivers were tested for statistical association with churn using Chi-square tests (α=0.05) in Excel, with contingency tables and revenue calculations cross-validated across SQL, Excel (SUMIF/SUMIFS/XLOOKUP/Pivot Tables), and Power BI (DAX).

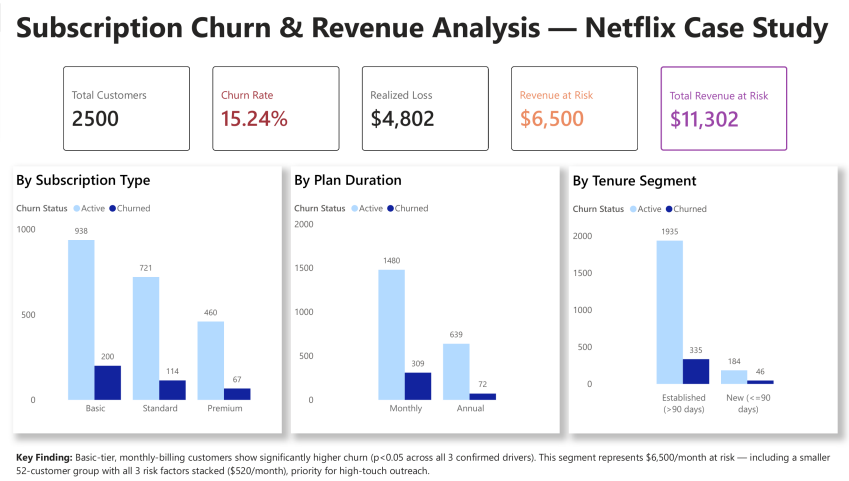
Key Findings

Factor	Comparison	p-value	Result
Plan Duration	Monthly 17.3% vs Annual 10.1%	0.000007	Significant
Subscription Type	Basic 17.6% vs Premium 12.7%	0.0109	Significant
Tenure Segment	New (<90d) 20.0% vs Established 14.8%	0.035	Significant
Country / Age / Gender / Device	No meaningful variation	> 0.36 (all)	Not significant

Revenue Impact

Realized loss (already churned)	$4,802 / mo
Revenue at risk — Basic + Monthly, still active (650 customers)	$6,500 / mo
of which: also New tenure (<90d) — all 3 risk factors stack (52 customers)	$520 / mo
Total revenue at risk	**$11,302 / mo (~$135,600 / yr)**

Dashboard



Recommendations

1. **Strengthen 90-day onboarding.** New customers (<90 days) churn 35% more than established ones (20.0% vs 14.8%). Introduce an onboarding email series, in-app tutorial, and check-ins at day 30/60/90.
2. **Incentivize the shift from Monthly to Annual billing.** This is the strongest driver identified (p<0.00001) — Monthly churn is nearly 2x Annual. Offer a 10–15% discount for annual commitment, prioritized for Basic-tier customers.
3. **Run a two-tier retention push.** The broad Basic + Monthly segment (650 customers, $6,500/month) warrants a mass retention campaign (email, in-app offers). Within it, a much smaller group of 52 customers stacks all three confirmed risk factors (Basic + Monthly + New tenure, $520/month) — small enough for high-touch, personal outreach (e.g., a direct call) as the highest-urgency priority.

Note: This project uses a public proxy dataset for subscription behavior modeling, as customer-level data for real subscription businesses is not publicly available. The business context is grounded in Netflix's real Q1 2022 subscriber-loss event.